

Where every new part goes

Four views of the stock coach, with the new hardware placed in it: the floor plan, a side section, the underbody and the roof. Each new part has a number that matches the schedule below, which says where it goes, what it takes the place of, how it mounts and what it connects to. Positions are inches forward of the rear wall. They come from the model and get confirmed with a tape measure.

INSIDE THE CABIN · 3 CHANGES - RV5 hub, disconnect and PV breakers go in the factory desk base, roadside rear corner under the lift bed (1, 3) - Epad on the wall at the desk (4) - Transfer switch and surge protector in the pantry base (11, 12) - Washer in the curbside corner at the forward end of the lift bed, against the pantry (13)	ROOF - Six panels in one crosswise row, split around the AC (6) - Combiner and cable gland (7, 5) - Chill Cube swaps into the existing opening (8) - Starlink Mini beside it (9)	UNDERNEATH AND CAB - Sealed box with both packs under the tail (2) - Two Orions under the passenger swivel seat (10)	NOT CONFIRMED YET - Height under the lowered bed at the washer corner (needs at least 33.7 in) - The other fit checks listed below
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Bluetti / 48 V DC

Solar

Alternator charging

120 V AC

Washer-dryer

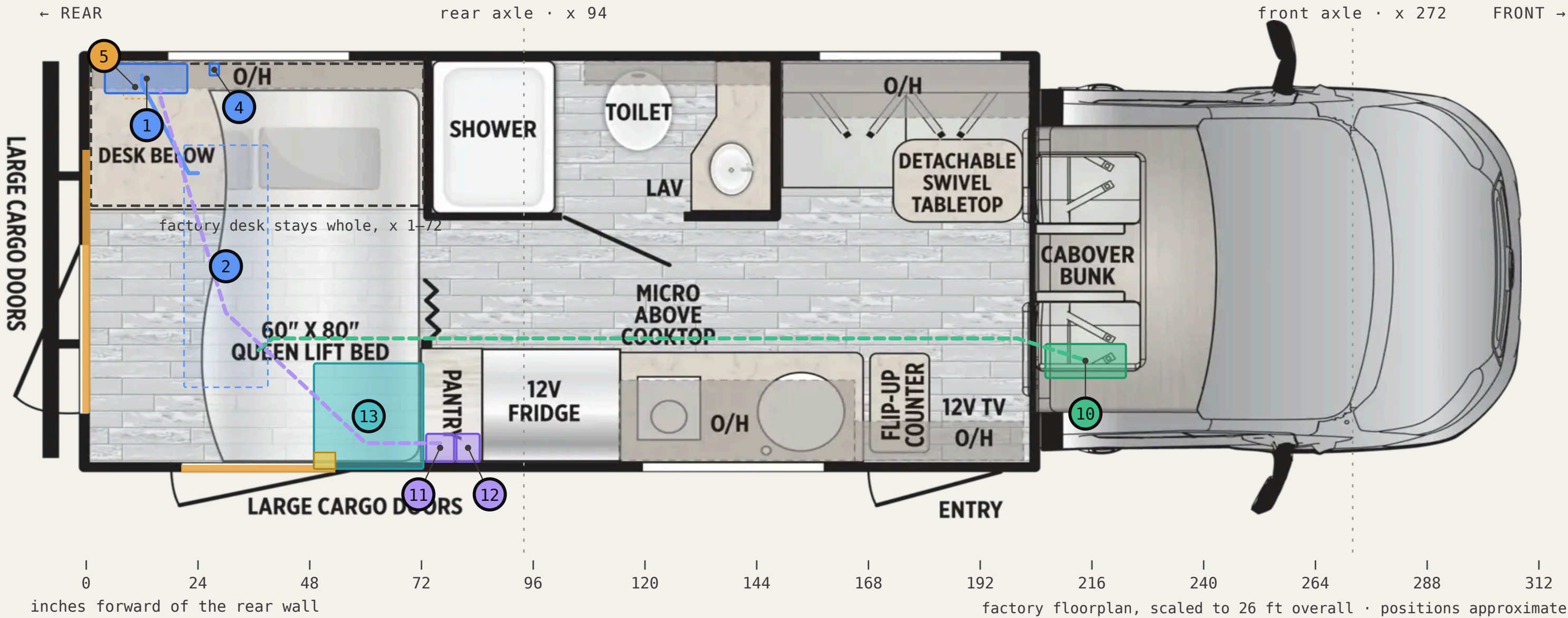
Stock coach

Fit check

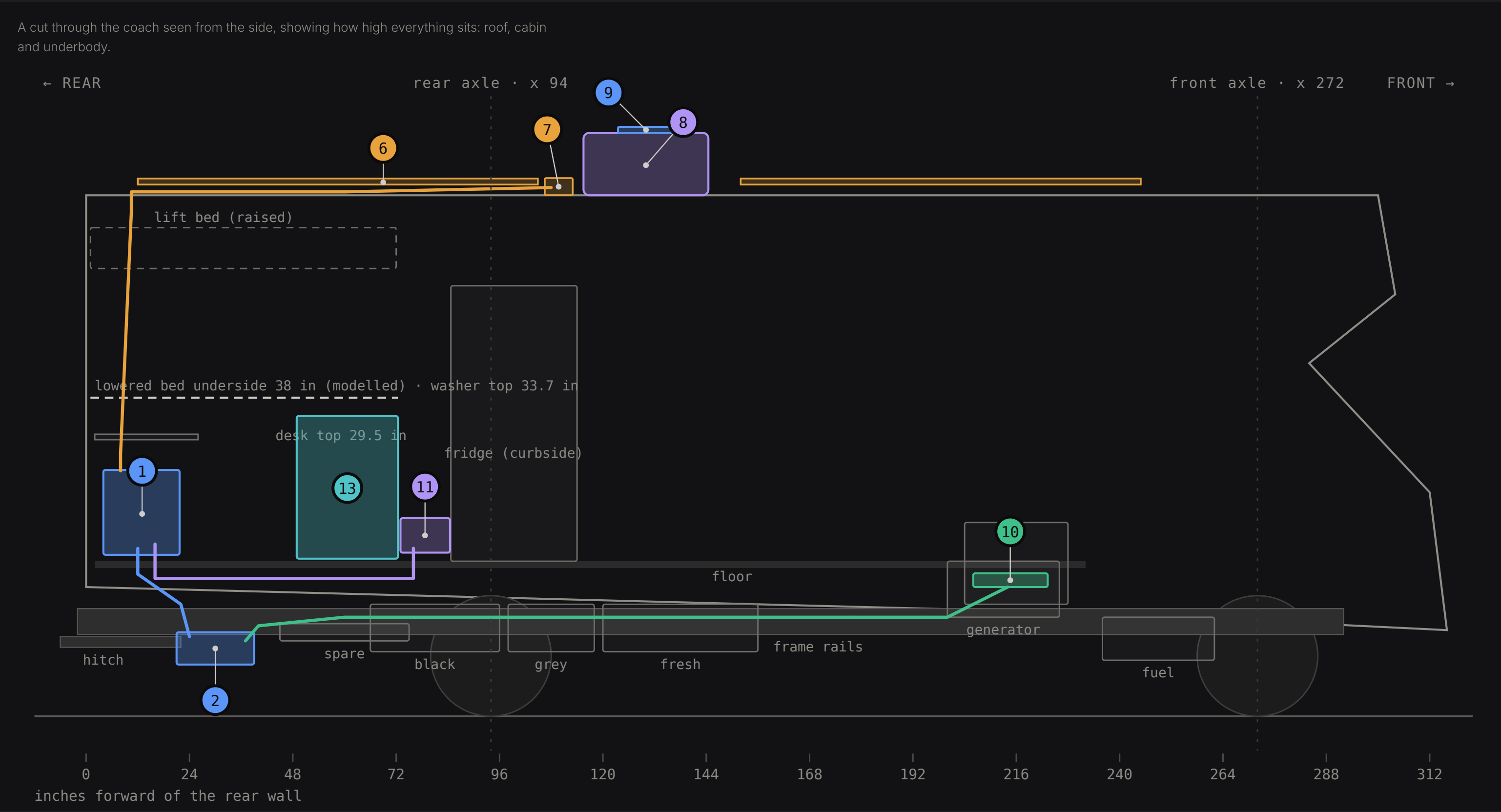
VIEW 1

Floor plan

East to West's factory 22NF floorplan, scaled to 26 ft, with the new hardware placed on it. Roadside (driver side) is at the top. Dashed shapes are above the ceiling or below the floor. Orange lines are the cargo door openings read off the floorplan.



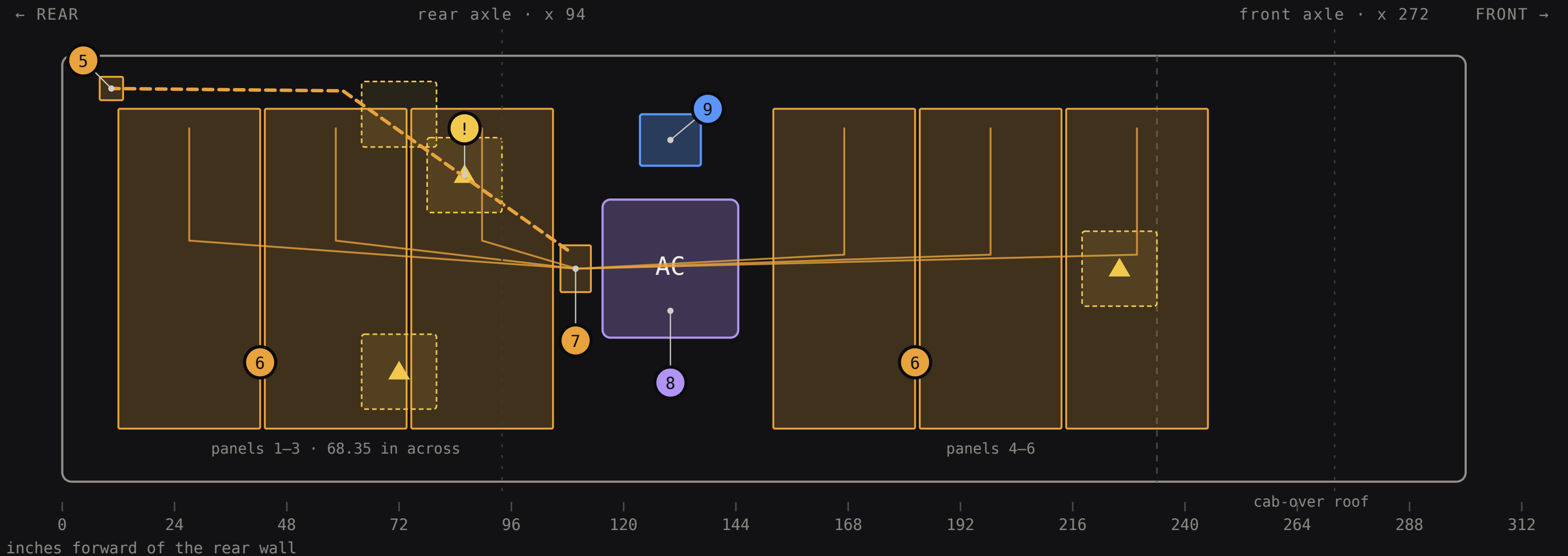
Side section



Underbody



Roof



PLACEMENT SCHEDULE

Every new part, where it goes and what it connects to

#	PART · WHERE IT GOES	POSITION	TAKES THE PLACE OF	MOUNT AND ACCESS	CONNECTS TO
1	Bluetti RV5 hub In the rear-end pedestal of the factory desk ("desk below"), roadside rear corner under the lift bed, on the floor against the roadside wall.	x 1–22 in from rear wall · roadside wall · floor level	The rear-end desk pedestal's storage. The rest of the desk is untouched.	Bolted to the floor through its mounting holes. Pedestal needs a vented front panel so the RV5 keeps 7.87 in of open air at its vents. Front panel removable for service.	48 V down through the floor to the packs (3 ft) · PV trunks from the roof gland straight above (14 ft each) · 120 V in from the transfer switch in the pantry base (about 9 ft) · 120 V out to the factory breaker panel (location to find) · 12 V out to the coach 12 V panel.

#	PART · WHERE IT GOES	POSITION	TAKES THE PLACE OF	MOUNT AND ACCESS	CONNECTS TO
2	2 × B4810 in a sealed box Under the tail, across the frame rails, between the hitch crossmember and the spare tire.	x 21–39 in from rear wall (about 64 in behind the rear axle) · 52 × 18 × 7.5 in box · bottom about 12 in off the ground	Nothing inside. Uses empty space behind the spare. Must clear the hitch receiver tube, the exhaust tip, and the dump valves.	Hangers bolted to the coach builder’s frame extension (not welded). Lockable lid, baffled vent, low drain, skid plate. Packs fused inside the box at their terminals.	48 V up through the floor into the desk base (3 ft) · alternator 48 V run from the cab (about 16 ft).
3	Main DC disconnect, PV breakers, pack fuse Disconnect and both PV breakers inside the desk base next to the RV5. The pack fuse goes in the battery box.	Desk base, beside #1 · pack fuse in #2	Shares the desk base with the RV5.	Reachable from inside the coach without crawling under. (The 3D model draws the disconnect on the box; this map moves it inside.)	Sits between the packs and the RV5, and between each PV trunk and its RV5 input.
4	Bluetti Epad (control screen) On the roadside wall at the desk, just above the desk base.	x 26–28 in · roadside wall · eye level when seated	Wall space only.	Surface mount. Ethernet to the RV5 directly below.	Data cable to #1 (under 3 ft).
5	Roof cable gland Rear roadside corner of the roof, directly above the desk.	x 8–13 in · 36–41 in roadside of centre	Nothing. New sealed penetration (or the factory solar side-port if it lands here).	Sealed gland with written roof-warranty coverage. The two PV trunks drop inside the rear roadside wall into the desk base.	Both PV trunks from #7 to #1.
6	6 × Callsun 275 W panels Roof, one crosswise row of six: panels 1–3 behind the AC, panels 4–6 ahead of it, running onto the cab-over roof.	x 12–105 and 152–245 in · each 68.35 in across the roof	Roof space only. The existing fans and skylight sit under the panels (see fit check).	Z-brackets or a low rail, about 2.5 in off the roof. Walk margin left on both sides.	Panel leads to #7 (4–12 ft each, about 48 ft total).
7	Combiner box (3 + 3, 25 A per panel) Roof centreline, between panels 1–3 and the AC.	x 106–113 in · centreline	Nothing.	Bolted to the roof, sealed. Two groups of three, each with a 25 A fuse per panel.	Panels in; two PV trunks out to #5 then #1 (14 ft each).
8	Furrrion Chill Cube 18K (ducted) The existing 14 × 14 in AC opening. Swaps one-for-one with the factory unit.	Opening modelled at x 116–145 in (centre x 130). Real position unverified.	The factory AC (removed). Its ceiling assembly is replaced by the Chill Cube air distribution box, which feeds the existing ducts.	Bolts through the existing opening. Confirm the air distribution box mates to the factory ducts before buying.	120 V from the main panel (RV5 output).
9	Starlink Mini Roof, roadside of the AC on a flat mount.	x 124–137 in · 22–33 in roadside of centre	Roof space only.	Flat mount, adhesive or bolted. Cable rides the PV gland or the AC chase.	12 V from the RV5 12 V output.
10	2 × Victron Orion-Tr 12/48 Under the passenger (curbside) cab seat, close to the starter battery.	x 206–223 in · curbside · under the seat base	Space under the passenger seat. The cab seats swivel to face the table, so the swivel base must leave room (fit check).	Screwed to the seat base or the floor beside it. Ignition-switched. Fuse at the starter battery.	12 V from the starter battery (about 3 ft) · 48 V back to the packs along the curbside rail (about 16 ft, about 16 A, so a small cable).
11	Transfer switch In the base of the pantry, curbside, just forward of the lift bed. Candidate spot: move it next to the factory power center if that turns out to be nearby.	x 73–79 in · curbside · floor of the pantry	The bottom shelf of the pantry.	Screwed to the floor. Service through the pantry door.	Generator in (length depends on the generator bay, unverified) · shore in through #12 · out to the RV5 AC input (about 9 ft).
12	Surge protector / EMS (hardwired) Beside the transfer switch in the pantry base.	x 79–85 in · curbside	The bottom shelf of the pantry.	Screwed to the floor next to #11.	Shore inlet in (inlet location unverified) · out to #11.
13	Splendide WDV2200XCD (vented) Curbside corner at the forward end of the lift bed, backed against the curbside wall with its side against the pantry wall.	x 48.9–72.4 in · y 20.9–43.5 in curbside of centre · footprint 23½ W × 22⅝ D × 33⅙ H in (plus about 0.6 in plinth) · centre about 33 in behind the rear axle	Garage floor in that corner, under the forward end of the bed.	Bracket to the floor. Fits under the lowered bed only if the bed’s underside sits at least 33.7 in above the floor (modelled 38 in, unpublished). Vent out the curbside wall. Its rear 4½ in overlaps the forward end of the curbside cargo door opening (x 20–54), leaving about 29 in of the opening clear.	Hot and cold cross under the aisle from the bath (shower directly opposite, about 4 ft) or come back from the galley sink. Drain to the grey tank, which the model puts under the floor just ahead of this corner. Own 15 A circuit from the main panel.

Cable lengths are measured along the routes drawn in the 3D model, rounded to the nearest foot.

What the six-panel row needs, and what to do if it doesn't fit

LENGTH THE ROW NEEDS 233 in Six panels at 30.3 in, 1 in gaps, 10.5 in for the combiner before the AC, 29 in of AC, and 7.5 in after it. In the model it runs from x 12 to x 245.	ROOF THE MODEL USED ~300 in, estimate The model treats the roof as flat from the rear cap over the cab-over. The row starts 12 in from the rear cap and ends 11 in past the house-to-cab-over seam (x 234). The 1 in margin quoted earlier compared the row with the house section alone. Measure both sections.	MAXXFAN UNDER A PANEL No A MaxxFan Deluxe with rain dome stands 5.0 in tall closed and 9.1 in open. Panels on 2.5 in brackets can't pass over it.
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FALLBACKS, IN ORDER

- Tall rails over the fans.** Raise the panels so their underside sits about 10 in off the roof, clearing an open MaxxFan. Panel tops end up about 11.4 in high, still under the Chill Cube's 14.5 in, so the coach's overall height doesn't change. The cost is a heavier rack and more wind load.
- Leave a fan open.** Shift the row so a fan falls in a gap. Each uncovered fan costs one panel slot, leaving five panels: 1,375 W wired 3 + 2. Two panels in parallel need no per-panel fuses.
- Short roof.** Drop the front panel: five panels need about 202 in. Four panels (2 + 2, 1,100 W) need about 171 in.
- Tighten the gaps.** Move the combiner off the row to the roof edge to save about 6 in.

REAR AXLE CHECK

Axle loads added by the build, against the Transit 350 ratings

The coach's empty axle weights aren't published, so the table works backwards. It totals everything the build and a loaded trip add to each axle, then shows the heaviest the *empty* axle can weigh and still stay under its rating (front 4,630 lb, rear 7,275 lb). Included: the build at its modelled positions, a full fresh tank (290.5 lb at the tank), a 40 lb tow bar on the receiver, a 200 lb driver, and 350 lb of gear spread midship.

SCENARIO	ADDED TO FRONT AXLE	ADDED TO REAR AXLE	EMPTY REAR AXLE CAN BE AT MOST	EMPTY FRONT AXLE CAN BE AT MOST
Panels 29.8 lb each · gear midship	+235 lb	+1344 lb	5,931 lb	4,395 lb
Panels 29.8 lb each · gear in the rear garage	+58 lb	+1521 lb	5,754 lb	4,572 lb
Panels 65.7 lb each · gear midship	+277 lb	+1518 lb	5,757 lb	4,353 lb
Panels 65.7 lb each · gear in the rear garage	+100 lb	+1695 lb	5,580 lb	4,530 lb

- What it means.** With an empty coach around 9,000 lb, the rear axle stays legal if the empty rear axle weighs no more than about 5,580–5,930 lb, or roughly 62–66% of the empty weight on the rear. The front axle has room in every case. A CAT scale reading of the empty coach answers it.
- Gear placement matters as much as the box.** Putting the 350 lb of gear in the rear garage instead of midship adds another 177 lb to the rear axle. Every 100 lb moved from the garage to midship takes about 50 lb off the rear.

- **Should the box move forward?** Moving the packs from 64 in behind the axle to directly over it takes 82 lb off the rear axle. In the model, the space between the rails at the axle is taken by the tanks. Keep the tail box unless the scale shows the empty rear axle above about 5,500 lb. If it does, move gear forward first, then look at the box.

BATTERY CABLE

The 48 V run from the packs to the RV5 is about 3 ft

The box sits under the rear floor, and the RV5 sits in the desk pedestal right above it and slightly behind. The modelled run goes down through the pedestal floor and into the box: 37 in, about 3 ft. The 10–15 ft guess came from earlier layouts with the hub further forward. The longest 48 V run is now the alternator line from the Orions: about 20 ft at about 16 A, so a small cable. The pack fuse stays at the pack terminals inside the box.

FIT CHECKS

Nine things the tape measure has to confirm

1. **Roof fans and skylight under the panels.** The panels sit 2.5 in off the roof, and the model puts two roof fans and the bath skylight under them. A MaxxFan Deluxe with rain dome is 23.2 × 16.6 in and stands 5.0 in tall closed and 9.1 in with the lid open (retailer spec, overall unit height). It cannot sit under a panel on 2.5 in brackets. See the roof fallback section.
2. **Roof length and AC position.** The row of six spans 233 in from the first panel edge to the last (x 12 to 245). The model treats the roof as flat from the rear cap over the cab-over to about x 300. Measure the flat roof length, whether the cab-over roof is flat and walkable, and the AC opening’s distance from the rear cap.
3. **Tail box vs. hitch and spare.** In the model the box starts right where the receiver tube ends. Measure the gap from the hitch crossmember to the spare, frame-extension height and bolt points, and exhaust-tip position.
4. **Desk base size.** The RV5 is 17.7 × 19.7 × 6.3 in and wants 7.87 in of air at its vents. Measure the factory desk base in the roadside rear corner to confirm it can take the RV5 with a vented front.
5. **Passenger seat swivel.** The cab seats swivel to face the table. Check what’s under the passenger swivel base and whether two Orions (each about 7.3 × 5.1 × 3.1 in) fit without blocking the swivel. Fallback: the floor beside the seat base or the pantry base with the transfer switch.
6. **Factory power center and generator bay.** Find the factory breaker panel and converter, and the generator bay. The transfer switch goes in the pantry base unless the power center is close by.
7. **Shore inlet location.** The model guesses the inlet is at the rear roadside. If it’s forward, the shore run to the transfer switch gets shorter.
8. **Cargo doors.** The factory floorplan shows large cargo doors in the rear wall and on the curbside under the bed. The curbside door swings outward, so it still opens; the washer takes about 4½ in off the forward end of its roughly 33 in opening. Measure the opening.
9. **Washer corner.** Measure the lift bed’s underside fully lowered at the forward curbside corner (needs at least 33.7 in), the floor space from the pantry wall back (needs 23½ in) and out from the curbside wall (needs 22⅝ in), where the bath and galley supply lines run, and where the grey tank inlet is.

3D REFERENCE

The same layout in the rendered model

